



OPEN Machine learning and molecular dynamics simulations aided insights into condensate ring formation in laser spot welding

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Condensate ring formation can be used as a benchmark in welding processes to assess the efficiency and quality of the weld. Condensate formation is critical as the resulting condensate settles into the powder thereby altering the quality of unconsolidated powder. This study investigates the intricate relationship between alloy composition, vapor pressure, and condensate ring thickness as seen in a two-dimensional micrograph. To study the process, laser spot welding was performed on 9 different alloys, and the inner spot weld diameter along with the condensate ring formation was studied. Leveraging machine learning models, experimental observations, and molecular dynamics simulations, we explore the fundamental factors governing condensate ring formation. The models, adept at predicting weld spot diameter and condensate ring thickness, identify laser power as a primary determinant for weld spot diameter followed by physical properties like hardness and density. Conversely, for condensate ring thickness, vapor pressure and melting point descriptors consistently emerge as paramount, as validated across all models. Molecular dynamics simulations on Ni-Cr alloys elucidate the vaporization dynamics, confirming the role of vapor pressure in governing surface vaporization. Our findings underscore the pivotal influence of vapor pressure and melting point descriptors in condensate ring formation. The convergence of machine learning predictions and simulation insights elucidates the dominance of these descriptors, offering crucial insights into alloy design strategies to minimize condensate ring formation in laser welding processes.

Keywords Laser spot welding, Laser spatter, Condensate ring, Machine learning, Molecular dynamics

Laser welding remains a cornerstone of manufacturing and construction industries, enabling the fabrication of a vast array of structures and components. From automotive and aerospace to shipbuilding and civil engineering, the versatility and strength provided by welding are indispensable^{1–4}. Although laser welding offers high precision and deep penetration, understanding the intricate details of laser-material interactions, such as the formation of condensate, remains a critical challenge. Minimizing the formation of condensates is critical as the condensate absorbs a portion of the incident laser energy, occluding the beam and reducing the effective power delivered to the powder bed⁵. It also causes beam scattering, increasing the size of the beam. This results in more sintered-on powder particles, or “satellites,” which are not fully melted into the dense strand material, producing poorly consolidated solid strands⁵.

Formation of condensates arises from impingement of a high intensity laser beam on a metal surface that induces a series of complex but intriguing physical phenomena. These processes occur simultaneously during and beyond the lifetime of the exposure of surface to the incident laser beam. Beam intensities of the order 10^6 W/cm²⁶ cause the top layer of most metals to melt and form a pool of molten metal. At such high powers, rapid metal evaporation takes place from the top layer of the melt pool and the resulting metal vapor exerts a downward pressure on the melt pool⁷. This pressure causes the top liquid layer to dip downwards until it takes the shape of the cone and forms a capillary with a vapor cavity⁸. Ref⁹ calculated the expression for recoil pressure as

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$$p_{\text{recoil}} = \frac{\text{power}_{\text{evaporation}}}{A_{\text{laser}} H_V} \sqrt{\frac{\pi k_B T_s}{2m_v}} \quad (1)$$

Where $\text{power}_{\text{evaporation}}$ is the laser power absorbed that was consumed for evaporation of molten metal, A_{laser} is the area of the incident laser beam, H_V is the enthalpy of evaporation, T_s is the surface temperature and m_v is the molecular mass of the vapor molecule. The p_{recoil} causes instabilities in the melt pool by entrainment of particles with the vapor stream, leading to the production of laser spatter¹⁰. The inception of a spatter occurs in a three-dimensional scenario, when a small volume (submicron) of molten liquid acquires sufficient kinetic energy in the direction perpendicular to melt pool surface, so as to detach and break free from the surface. More precisely, the condition can be mathematically expressed as⁷

$$\rho v_z^2 > \frac{2\sigma}{R} \quad (2)$$

Where, ρ is the density of the molten metal, v_z is the vertical component of the velocity of the escaping droplet, σ is the surface tension and R is the local radius of curvature of the melt pool. Under most circumstances, spatter is produced when a melt pool has low boiling point elements that result in explosive ejections due to superheating combined with ablation pressure⁷. This ejected material solidifies into nanoparticles referred to as condensate¹⁰, which typically line the interior of a build chamber in laser powder bed fusion (L-PBF) processes. Both the laser spatter and condensate collectively constitute what is termed heat-affected powder, originating from high-temperature conditions. Differentiating between the two is primarily based on size: laser spatter aligns closely with the powder's size, while condensate particles are submicron in size. Ref⁴¹. conducted a study characterizing spatter from 316 L stainless steel, Al-Si10-Mg, and Ti6Al-4 V produced during L-PBF. They found that laser spatter particles were larger than the original powder particles and exhibited oxidation on their outer surfaces. In an intriguing study by Liu et al.¹², the impact of spatter formation on final part properties was investigated by using unscreened laser spatter to build a part. The research revealed a decrease in part strength and ductility due to contaminants present in the laser spatter. Both studies highlighted the potential for laser spatter to settle into the powder bed, compromising the reusability of unconsolidated powder due to its differing properties compared to the virgin material.

While spatter has been extensively studied in the past few decades^{13–19}, the condensate production during laser melting has been only sparsely investigated. Condensate particles are formed when hot metal vapor evaporating from the top surface of the liquid pool condenses on cooler areas of the apparatus forming a mass of tiny metal nanoparticles. One of the first efforts to quantify the size of the condensate particles was carried out by²⁰ where the authors carried out a spectroscopy of the weld plume and used the multi-wavelength method of particle sizing to calculate the size of the condensate particles. They reported the condensate particles to have a size between 80 and 100 nm. Sutton et al.¹⁰ carried out an investigation on spatter and condensates using the laser powder-bed fusion (L-PBF) technique on 304 L stainless steel, and reverified that the condensate particles are sized less than 100 nm and stick on to the surfaces of virgin powder as masses of nanoparticles. They also found that the condensates are rich in oxygen, carbon and silicon which can considerably influence the part quality and properties.

Since condensate formation depends on factors like the melting point of elements in the melt pool and the overall thermal conductivity of the metal/ally, its formation can be related to the chemistry and other physical parameters of the system like heat capacity, thermal conductivity, dynamic viscosity, melt density, melt temperature, latent heat of fusion and evaporation and mass of the vapor molecule²¹. Currently there are no works that estimate the amount of condensate generated during a laser melting process given the complexities involved in modeling a multiphase process that encapsulates all three phases (solid metal, liquid metal and metal vapor). Given the high computational cost of mechanistic modeling, machine learning (ML)^{22–25} can prove to be a viable technique to study the relationship between material chemistry and condensate ring formation. ML has previously been used to predict defect formation in laser-assisted powder bed fusion (L-PBF) and its dependence on mechanistic variables like Marangoni number, energy density, surface tension, Richardson number²⁶ or alloy chemistry²⁷, where the authors collected required data from literature about what conditions and process parameters induced the balling defect in L-PBF.

Few past works have however attempted to model the spatter formation although not the condensate formation itself. Ly et al.²⁸ investigated the dynamics of metal micro-droplet ejection in metal additive manufacturing (AM), specifically focusing on the laser powder bed fusion (LPBF) process. Ultra-high-speed imaging of melt pool dynamics revealed that the dominant mechanism for micro-droplet ejection is not the laser-induced recoil pressure, as commonly believed in laser welding. Instead, the ejection is primarily driven by vapor-driven entrainment of micro-particles facilitated by an ambient gas flow. Simulations of the laser-powder bed interactions support the experimental results, describing the physics of droplet ejection under strong evaporative flow. Another work²¹ simulated deep penetration laser beam welding using the Smoothed Particle Hydrodynamics (SPH) method, modeling solid and liquid phase behaviors, heat transfer, and phase transitions. It introduced an improved recoil pressure model based on absorbed irradiance distribution, validated through comparison with traditional temperature-dependent models. Key findings show that recoil pressure significantly affects capillary formation and melt dynamics. High recoil pressures, generated at high irradiance points, support local capillary formation while surface tension causes temporary collapses, resulting in a narrow, deep melt pool.

The aforementioned studies are among the few that have attempted to model the spatter formation. However, there is a noticeable gap in the literature, as no specific works have been found that directly model the formation of condensate rings during laser melting.

In the present work, ML has been leveraged to predict the formation of a condensate ring and its thickness, using a large set of experimental data on various metals in a range of laser powers. Single pulse laser shots on metal plates were utilized for their ability to produce a weld spot with a surrounding condensate ring. Laser spot weld images were collected from a variety of metal alloy plate materials such as Mg, 316 stainless steel, and a variety of Ni-based superalloys. Images of the weld shots and condensate rings were then used to obtain the size of the spot weld and condensate ring diameters by using an image processing algorithm. The image data for all 800 images then form our training set to develop a ML model for the prediction of condensate ring thickness. In addition to predicting the condensate ring thickness, the model also reveals the contribution of various descriptors like vapor pressure, thermal conductivity and latent heat of evaporation & fusion, among others, in determining the condensate ring thickness. Specifically, the models collectively identify vapor pressure of the material as significant parameter in governing the formation of the condensate ring. This finding is confirmed further by molecular dynamics simulations and experimental observation using SEM.

Methods

Sample preparation

Stainless steel (316) samples were obtained from McMaster-Carr and all other metals from Good Fellow. Nickel chrome alloys were obtained from SAES Getters. Each sample set was compared to metal with the same processing parameters and if possible, within the same lot to minimize sampling error. The samples employed were either 4 mm thick bar stock (2.54 cm per side sections) or shim stock with a 500 μm thickness and the same lateral dimensions, depending on experiment.

The samples were mirror finished (± 10 nm roughness R_a basis) with in-house equipment and potted in polystyrene epoxy for later ease of use, sample control, and standardization of surface heights. The “pucks” of potted metal were then issued unique identification numbers to maintain control of the sample and to link information to processing parameters.

Laser shots

The industrial laser used in this study was a IPG photonics ytterbium laser Model YLR-300/3000-QCW-MM-AC-Y12, with a 1064–1070 nm emission wavelength range, a focal distance of 6 cm, a power range from 100 to 3000 W, and an emission time from 1 to 50 ms. The optical setup was enclosed in an argon environment and kept below 10 ppm ambient gases during use. The vendor-supplied software was employed to fire the laser and control the emission parameters. The emission pulse was set to a 5 ms single pulses with 1 s between each shot to allow the metal in the local region to cool. The laser had a Gaussian beam profile, with a beam divergence parameter of 1.6 mrad and a spot size of 92.8 μm diameter. The shot power employed in this study varied from 108 to 600 W (experimentally determined, the input values were 150–610 W). Figure 1 shows the inside configuration of the laser and translation stage (Fig. 1a), as well as a still of the melt pool, weakly ionized plasma, and condensate development (Fig. 1b).

Microscopic images of laser shots

The optical data was recorded for each shot on an OMAX 2400 epifluoroscope set to a final (image) magnification of 500X. The data was recorded with a vendor supplied 18-megapixel camera mated to the microscope with a 0.5X reducing lens to increase the field of view. Two 100 W Fibreroptic halogen-bulb white light sources were used to illuminate the samples and were set to 60 degrees to the zenith and in-plane with the sample. The magnification

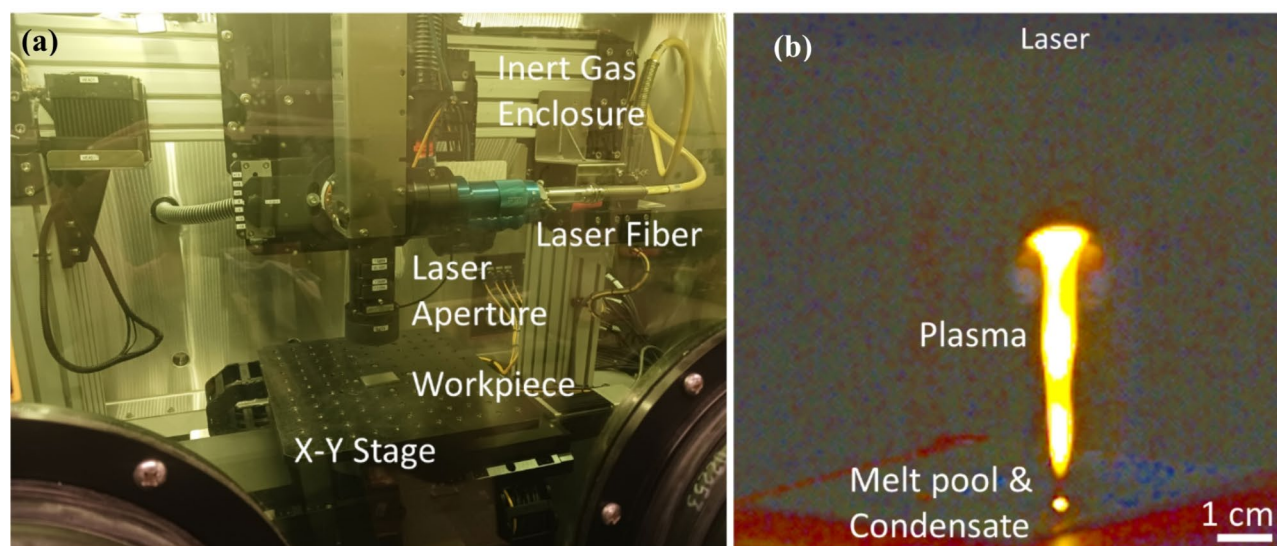


Fig. 1. (a) Photograph of the laser stage through the protective glass. (b) High-speed still of the melt pool and condensate glowing as well as the weakly ionized plasma above workpiece.

and orientation of the illumination was kept constants for all optical images to facilitate the creation of the data for the machine learning algorithm.

Scanning electron microscopy

Scanning electron microscopy was performed with the use of a JEOL IT800 field emission scanning electron microscope. EDS mapping was accomplished using an Oxford Instruments Ultim Max 170 mm² detector coupled with the AZtec Nanoanalysis software, v. 6.1 HF3. Secondary electron images were collected using a 5 kV accelerating voltage, 10.4 mm working distance and probe current setting of 75. Magnification varied from 80X to 5.5kX. Drift correction was enabled during multipass EDS mapping.

Image processing algorithm for diameter calculation

For identifying the diameters of the inner spot diameter and the outer condensate ring diameter, an image processing algorithm was used. The process of detecting circles using the Hough Circle Transform²⁹ in OpenCV with Python involves several sequential steps. Initially, the image is loaded and converted to grayscale to simplify subsequent operations. Preprocessing techniques like blurring or noise reduction are applied to enhance the image quality. The heart of the method relies on the Hough circle detection process. This algorithm detects circles within the preprocessed image by transforming it into a specialized parameter space. Peaks in this parameter space correspond to circles, with their parameters, such as center coordinates and radii, identified. After circle detection, the circles are visualized by drawing them on a copy of the original image. This step allows for a clear representation of the identified circular shapes against the image background, aiding in result interpretation. Finally, the resulting image containing the identified circles is displayed for analysis and interpretation. Adjusting detection parameters like minimum or maximum possible radius of circle in the image during the circle detection step is crucial for accurate identification of circles based on the specific characteristics of the image. The image processing algorithm is programmed to automatically run on all the images and the inner weld spot diameter and condensate ring thicknesses (as seen in a 2 dimensional micrograph) are tabulated. A total of around 800 images were used in this study. For all 9 alloys, 15 of the diameters from minimum to maximum at regular intervals, obtained from the image processing algorithm were compared with those obtained from manual measurements using ImageJ program. The ImageJ readings were within $\pm 5\%$ tolerance of that obtained from our image processing algorithm. The condensate ring thickness as seen on the two dimensional micrographs in Fig. 2 is obtained as (outer diameter – inner diameter)/2.

Dataset and feature engineering for prediction of inner spot weld diameter and outer condensate ring thickness

A series of alloys were selected for studying the weld spots and condensate ring. The alloys selected for study were MgAl, Ni, hot worked SS316, cold worked SS 316, IN 617, Hastelloy, Ni70Cr30, Ni80Cr20, and Ni95Cr5. The descriptors included in the ML model are listed in Table 1. Some essential descriptors like the laser power and chemical composition of the alloys were included as they are primal in determining the weld spot diameter and the formation of condensate ring. To capture more in-depth signatures of the spot-welding phenomenon, some thermal descriptors like melting point of the elements, thermal conductivity, specific heat, and density were included.

In several previous works, hardness and UTS have been associated with atomic bond strengths in materials^{23,30–34} and therefore have been included here as they can be good predictors of bond breakage during spot welding and melting. Another indicator of vaporization tendency of metal atoms from the surface during melting is the vapor pressure of the elements in the alloy. Vapor pressure represents the tendency of a material to transition from a solid or liquid phase to a vapor phase. During welding, the high heat from the laser can cause the localized melting of the alloy, leading to vaporization of some of its components. Elements with higher vapor pressures are more likely to evaporate and contribute to the formation of vapors in the weld pool. The vaporized material subsequently condenses upon cooling, forming condensate around the weld region. Elements with higher vapor pressures in their solid or liquid phases are more likely to contribute to this condensate formation due to their tendency to evaporate and then condense upon cooling. By considering vapor pressure as a descriptor in the predictive model, it helps differentiate between elements based on their propensity to vaporize and contribute to condensate formation, thereby aiding in understanding the relative impact of different alloy components on the resulting condensate ring thickness. The vapor pressure for the elements have been calculated using the Antoine equations³⁵ expressed by the equation $\log(P) = A + \frac{B}{T} + C \cdot \log(T) + D \cdot T \cdot 10^{-3}$, where A, B and C are constants for a material and T is the melting temperature. The coefficients for the equation have been taken from³⁶ and listed in Table 2.

Ideally, the vapor pressure should be calculated at the vaporization temperature, as vaporization is a highly out-of-equilibrium process that occurs when the saturated vapor pressure equals the ambient pressure. The vaporization temperature more accurately represents the conditions under which violent material ejection and rapid condensation occur only when the surface temperature surpasses the vaporization temperature, as noted in studies such as those by Hirano et al.³⁷ and Knight³⁸. However, after a thorough review of the available literature, we found that the most comprehensive source of vapor pressure coefficients for the elements relevant to our work (Mg, Al, Ni, Fe, Cr, Mo, Mn, Nb, Ti, Co, and W) was provided by Alcock et al.³⁶. In this dataset, vapor pressure equations and their corresponding coefficients are reported at the melting temperature for many of these elements. Unfortunately, the data required for calculating vapor pressures at the vaporization temperature are not available for several crucial elements in our alloy system, including Mg, Cr, Mo, Mn, Nb, and W. Given these limitations, we were constrained to use the available vapor pressure coefficients at the melting temperature. While this choice introduces an approximation in representing the vaporization process, we believe that it still provides meaningful insights, especially considering that localized regions within the melt pool during laser

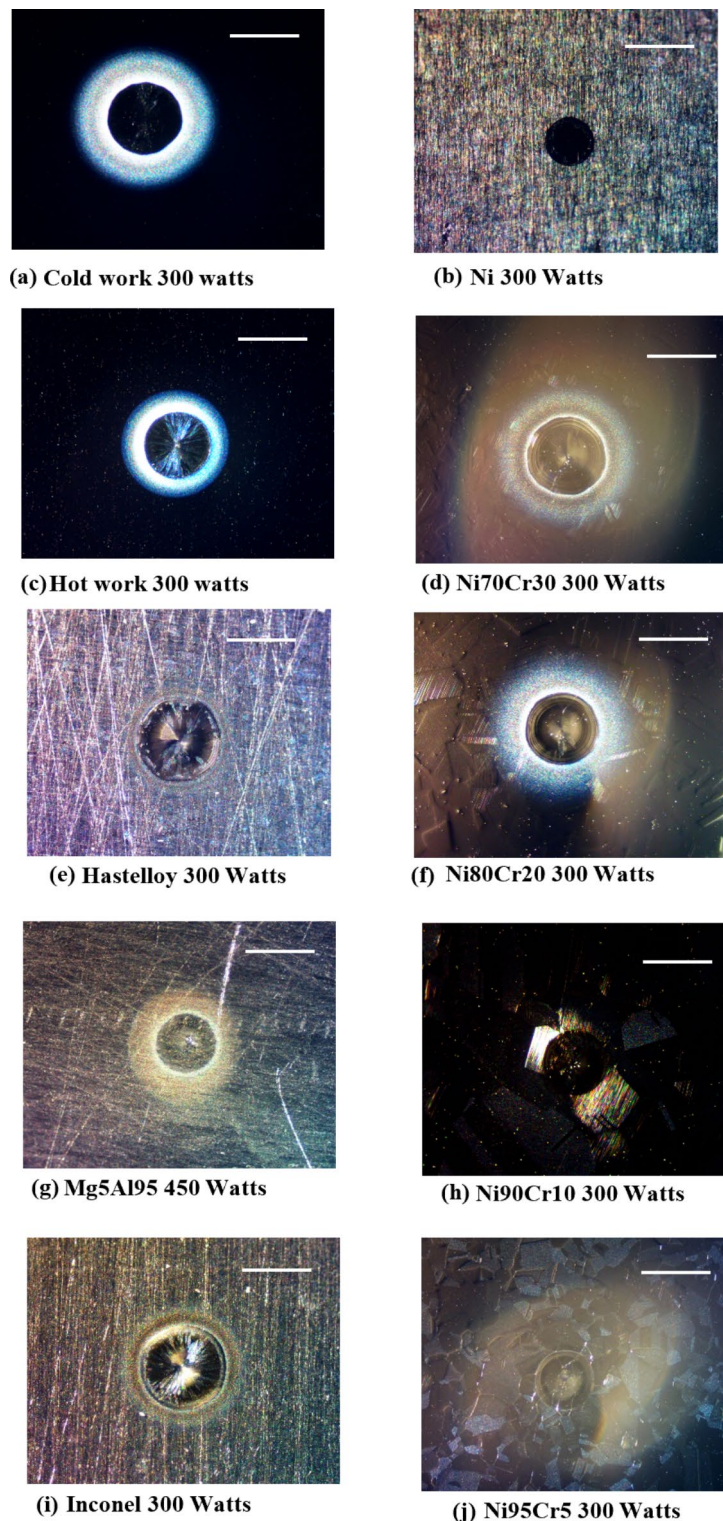


Fig. 2. Top view optical micrographs of laser weld spots on (a) SS 316 cold worked, (b) Ni, (c) SS 316 hot worked, (d) Ni70Cr30, (e) Hastelloy, (f) Ni80Cr20, (g) Mg5Al95, (h) Ni90Cr10, (i) Inconel and (j) Ni95Cr5 alloys. Scale bar in the inset denotes 415 μm .

welding may not uniformly reach the vaporization temperature. The melting temperature serves as a reasonable proxy for modeling vaporization behavior in regions where the material is transitioning between solid, liquid, and vapor phases under intense laser irradiation. By using melting temperature coefficients, the calculated vapor pressure may slightly underestimate the true vaporization behavior, leading to minor deviations in the predicted condensate ring thickness. Despite these limitations, our approach remains effective for capturing the

Feature	Shorthand	Description
Aluminum	Al	Metal present in alloy
Iron	Fe	Metal present in alloy
Nickel	Ni	Metal present in alloy
Other elements	...	9 other elements present in alloys in dataset
Minimum melting temperature	MinMelt	Minimum value of the melting temperature of metals present in alloy
Maximum melting temperature	MaxMelt	Maximum value of the melting temperature of metals present in alloy
Melting point	T_m	Melting point of the alloy as a whole
Thermal conductivity	Th. Con.	Minimum value of the thermal conductivity of metals present in alloy
Specific heat	Sp. heat	Minimum value of the specific heat of metals present in alloy
Density	ρ	Density of the alloy
Hardness	H	Hardness of the alloy
Ultimate tensile strength	UTS	Ultimate tensile strength of the alloy
Laser power	LaserPower	Laser wattage
Minimum vapor pressure	MinVP	Minimum value of the vapor pressure of metals present in alloy
Maximum vapor pressure	MaxVP	Maximum value of the vapor pressure of metals present in alloy
Average vapor pressure	Avg.VP	Average value of the vapor pressure of the alloy

Table 1. Descriptors used for training the models to predict Weld spot diameter and condensate ring thickness.

Element	A	B	C	D	T_{melt}
Al	9.459	− 17,342	− 0.7927	0	933
Mg	8.489	− 7813	− 0.8253	0	923
Ni	10.557	− 22,606	− 0.8717	0	1726
Fe	7.1	− 21,723	0.4536	− 0.5846	1808
Cr	6.8	− 20,733	0.4391	− 0.4094	2130
Mo	11.529	− 34,626	− 1.1331	0	2890
Mn	12.805	− 15,097	− 1.7896	0	1519
Nb	8.822	− 37,818	− 0.2575	0	2741
Ti	11.925	− 24,991	− 1.3376	0	1930
Co	10.976	− 22,576	− 1.028	0	1768
W	2.945	− 44,094	1.3677	0	3695

Table 2. Recommended formulations for the vapor pressures of the elements used in the study³⁶.

overall trends and interactions between the material properties and laser welding parameters. We have validated this model against experimental data and found that it provides reasonable agreement within acceptable error margins. The weighted average of the elements is taken to get the vapor pressure of the alloy similar to the methods adopted by the ML community for calculating properties like theoretical melting temperatures and density^{31,34,39,40}.

The interaction between high-irradiance laser light and molten metal is governed by complex multiphysics processes⁴¹. One of the critical parameters influencing these processes is absorptance - the proportion of incident laser energy absorbed by the material⁴². Absorptance is not a static property but varies dynamically during the laser welding process, influenced by factors such as surface roughness, surface temperature, laser wavelength, laser power, and material state (solid, liquid, or vapor). In particular, absorptance tends to fluctuate as the surface undergoes rapid changes due to thermal expansion, phase transitions, and vaporization. Measuring absorptance under these conditions, especially for molten metals, presents significant experimental challenges. The surface geometry constantly changes due to thermal expansion, and phenomena such as vapor ejection and surface turbulence introduce further complexity. Temperature measurements on molten or boiling surfaces are difficult because of non-homogeneous emissivity, and precise emissivity values are often unavailable at high temperatures. Absorptance can be measured using calorimetric (direct absorption) or radiometric (indirect absorption) methods⁴³, but these techniques are challenging to apply uniformly across a wide range of materials and conditions. Our recent work⁴⁴ investigated the impact of surface roughness on laser absorption. The study revealed considerable temporal variations in absorption, demonstrating that absorptance is highly sensitive to surface conditions that evolve during the laser-material interaction. This sensitivity highlights the complex nature of absorptance, especially when the material transitions through different phases (solid, liquid, vapor) and experiences surface phenomena such as vapor ejection and thermal expansion. In the present work, we are working with over 9 different alloys, with laser power ranging from 150 to 600 watts. Given the large number

of variables involved, it is impractical to experimentally measure absorbance for all combinations of material and process conditions. Consequently, we have opted not to include absorbance as a direct variable in our current modeling approach. Instead, we rely on laser power as a practical approximation for energy input, while acknowledging that this introduces a degree of approximation.

ML models

Machine learning models heavily rely on the characteristics and volume of available data. The confidence in their predictions requires acknowledgment of inherent uncertainties. Therefore, in such a scenario, it becomes imperative to train multiple models on the provided data and meticulously compare their outcomes. The objective is to ascertain which descriptors or features exert the most substantial influence on determining weld spot diameter and condensate ring thickness. This study involves employing a diverse array of regression models to comprehensively analyze the dataset and capture nuanced relationships between alloy composition descriptors and the formation of weld spot diameter and condensate ring thickness. The ensemble of models selected for this analysis includes:

- Linear Regression: A fundamental model that assumes a linear relationship between descriptors and diameters, offering a basic benchmark for predictions.
- Linear Regression with Polynomial features: This model transforms the input data into polynomial terms before fitting a linear regression model. The pipeline first generates polynomial features up to the specified degree (in this case, degree 2), scales them, and then applies linear regression on these polynomial features.
- CatBoost Regression: An advanced gradient boosting algorithm designed to handle categorical variables efficiently and optimize predictions.
- Gradient Boost, Adaboost, Decision Tree: Ensemble methods that iteratively build models to correct errors of predecessors, potentially capturing complex interactions among descriptors.
- Gaussian Process: A probabilistic regression model effective in capturing non-linear relationships and estimating prediction uncertainties.
- Ridge and Lasso Regression: Regularized linear regression methods aimed at reducing overfitting and identifying influential descriptors.

By employing this ensemble of models, each with its unique strengths and assumptions, the analysis aims to comprehensively explore and compare the relative importance of different alloy composition descriptors in determining the weld spot diameter and condensate ring thickness. The assessment involves examining model performance metrics, feature importance scores, and coefficients across various models to identify descriptors significantly contributing to the prediction of weld characteristics.

Molecular dynamics simulations

MD simulations are performed using the LAMMPS code and atomistic visualizations are done in OVITO. Laser spot welding on three alloys, Ni70Cr30, Ni80Cr20 and pure Ni were simulated under identical conditions to understand the role of chemistry and the influence of pressure on vaporization of atoms from the surface. The embedded-atom model (EAM) potential developed by Zhou et al.⁴⁵ is used to define the Ni-Ni, Cr-Cr and Ni-Cr interactions. The EAM is a many-body potential which is formulated as

$$E = F_a \sum_{j \neq i} \rho_i(R_{i,j}) + \frac{1}{2} \sum_{j \neq i} \phi_{\alpha, \beta}(R_{i,j}) \quad (3)$$

Where E is the total energy of the combined system, $R_{i,j}$ is the distance between j^{th} and i^{th} atoms, F is the embedding energy of the i^{th} atom, ρ is the electron density contribution, and ϕ is the short range pairwise potential energy, α and β being the element types of atoms i and j . The EAM potential has been previously validated and used for laser melting MD simulations and has been found to produce appreciable accuracy. In the present work, the density obtained for Ni80Cr20 is 8.30 gm/cc which is good agreement with the industrial specification of 8.4 gm/cc, and that for Ni is 8.85 gm/cc which also agrees well with the industrial specification of 8.90 gm/cc. A pure FCC crystal is obtained after equilibration for all 3 alloys, which corroborates the accuracy of the potential.

Laser melting simulations in MD

A $7 \text{ nm} \times 7 \text{ nm} \times 35 \text{ nm}$ cuboidal simulation cell of $\text{Ni}_x\text{Cr}_{(100-x)}$ was constructed as shown in Fig. 10a. A vacuum is maintained in the top and bottom 10.5 nm region, thereby creating two free surfaces at the top and bottom. Periodic boundary conditions are used in all three directions. The system was equilibrated in the NVE (microcanonical) ensemble for 100 ps. To simulate the laser beam striking the bulk, a $1 \text{ nm} \times 1 \text{ nm} \times 35 \text{ nm}$ cylindrical section was chosen to be used as the heat absorbing region by the laser beam as shown in Fig. 10f. This region was supplied with a non-translational kinetic energy of 1000 eV/femtosecond, which can be simply understood as heat added at every timestep. The magnitude of power supplied to the system is a few orders of magnitude in excess of 10^6 W/cm^2 , to ensure the production of spatter and accelerate the process analysis. Since vacuum was maintained at both top and bottom regions, there are two free surfaces. Analysis of the top free surface is done to study the atoms that vaporize into the vacuum under the action of the laser beam.

Results

The results from the optical microscopy are shown in Fig. 2a–j for the various materials evaluated. Imaged at the same magnification, these findings illustrate weld spot diameter and condensate ring formation are significantly

influenced by the material's chemistry and structure. Extremes are observed, such as negligible condensate ring formation in nickel (Ni) compared to a substantial weld spot diameter and thick condensate ring formation in cold-worked stainless steel 316 (SS 316). Our study involved the analysis of micrographs using a custom image processing algorithm to quantify both the weld spot and condensate ring diameters.

Figure 3 illustrates the outcomes of our image processing algorithm, precisely identifying and measuring the circles corresponding to the weld spot diameter and the condensate ring diameter. The algorithm employs a pixel-based calculation to determine diameters, subsequently converting the measurements from pixels to microns for accurate dimensional analysis. This approach allows for a precise assessment of the weld spot and condensate ring sizes, enabling a detailed understanding of the relationship between material chemistry, weld characteristics, and condensate formation.

Figure 4a and b present the comprehensive results detailing weld spot and condensate ring diameters for various alloys. Alloys that do not exhibit condensate ring formation at the optical measurement scale were omitted for brevity. Notably, Hastelloy demonstrates the largest spot diameter and the thickest condensate ring among the examined alloys, followed closely by Inconel. These alloys, predominantly comprised of Ni-Fe-Cr elements, showcase prominent weld spot sizes and substantial condensate ring thicknesses. In contrast, pure Nickel (Ni) exhibits the smallest spot diameter and no discernible condensate ring formation. These observations underscore the significant influence of material chemistry on condensate ring formation. The trends suggest that the underlying factors governing condensate ring formation at a given weld power are intricately linked to the alloy's elemental composition.

A comprehensive investigation of the Ni_xCr_y series was performed using SEM to elucidate the impact of varying elemental compositions on the morphology and characteristics of the condensate ring. The Ni_xCr_y series constituted the most substantial portion of the training dataset (discussed subsequently in Fig. 7), which boosts the model's capability to generate reliable predictions for a broad range of NiCr compositions. Therefore, an in-depth experimental analysis of NiCr alloys is imperative for corroborating the model's results, thereby enhancing its predictive accuracy and validating its applicability to diverse compositional variations. These findings from SEM observations are shown in Fig. 5 where Ni80Cr20 forms a thicker condensate ring than the Ni95Cr5. The condensate ring is most easily visualized in the 80X SEI image, as the bright white contrast around the perimeter of the spot weld. This halo effect is significantly less pronounced on the 5Cr sample. At higher magnifications the topography of this condensate layer becomes apparent (5.5kX). The redeposition of this layer was evaluated using EDS, where it was identified as a Cr & O compound. This finding illustrates that, despite the full spot weld going through a melt-resolidification process, a measurable level of de-alloying occurred such that Cr (previously bound within the base metal) was ejected disproportionately and allowed to resolidify on the surface and oxidize. The resultant oxide ring is sparse and patchy in the 5Cr sample while the 20Cr specimen exhibits considerably more redeposition. Evaluation of the Ni-Cr phase diagram brings clarity to this difference – increased Cr content lowers the liquidus temperature of the alloy. For the same time duration of the laser pulse, the increased Cr content material will experience more time at melt. The lower vapor pressure of the increased Cr material indicates higher Cr content will increase the extent of condensate ring formation. It is interesting that this effect becomes apparent over the time domain of a 5 ms pulsed spot weld.

Figure 6 depicts a heatmap illustrating the correlations between the descriptors utilized for predicting weld spot and condensate ring thickness in this study. The Pearson correlation coefficient, denoted by 'P', serves as a robust indicator of these correlations, where values close to +1 or -1 signify strong positive or negative correlations, respectively. The analysis reveals notable strong positive correlations between specific descriptor pairs, such as the atomic percentage (% at.) of Al with Mg, and Ti with Nb. However, it's crucial to note that these

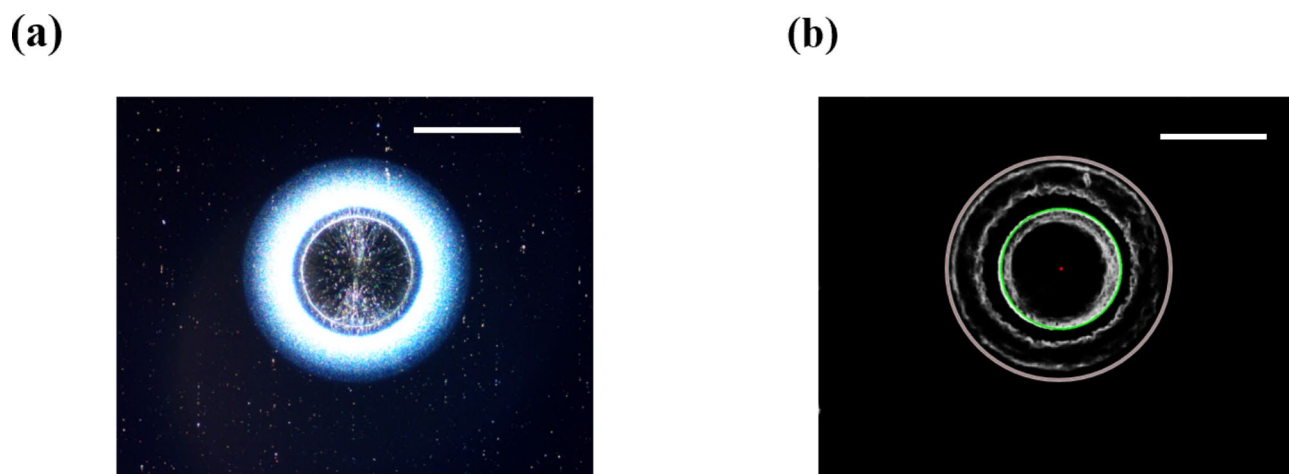


Fig. 3. The functioning of the image processing algorithm on a weld spot and condensate ring on cold worked SS 316. (a) The raw image of the weld spot is converted into (b) a grayscale image in which the inner and the outer periphery are identified as the weld spot diameter and the condensate ring diameter. Scale bar in the inset denotes 415 μm .

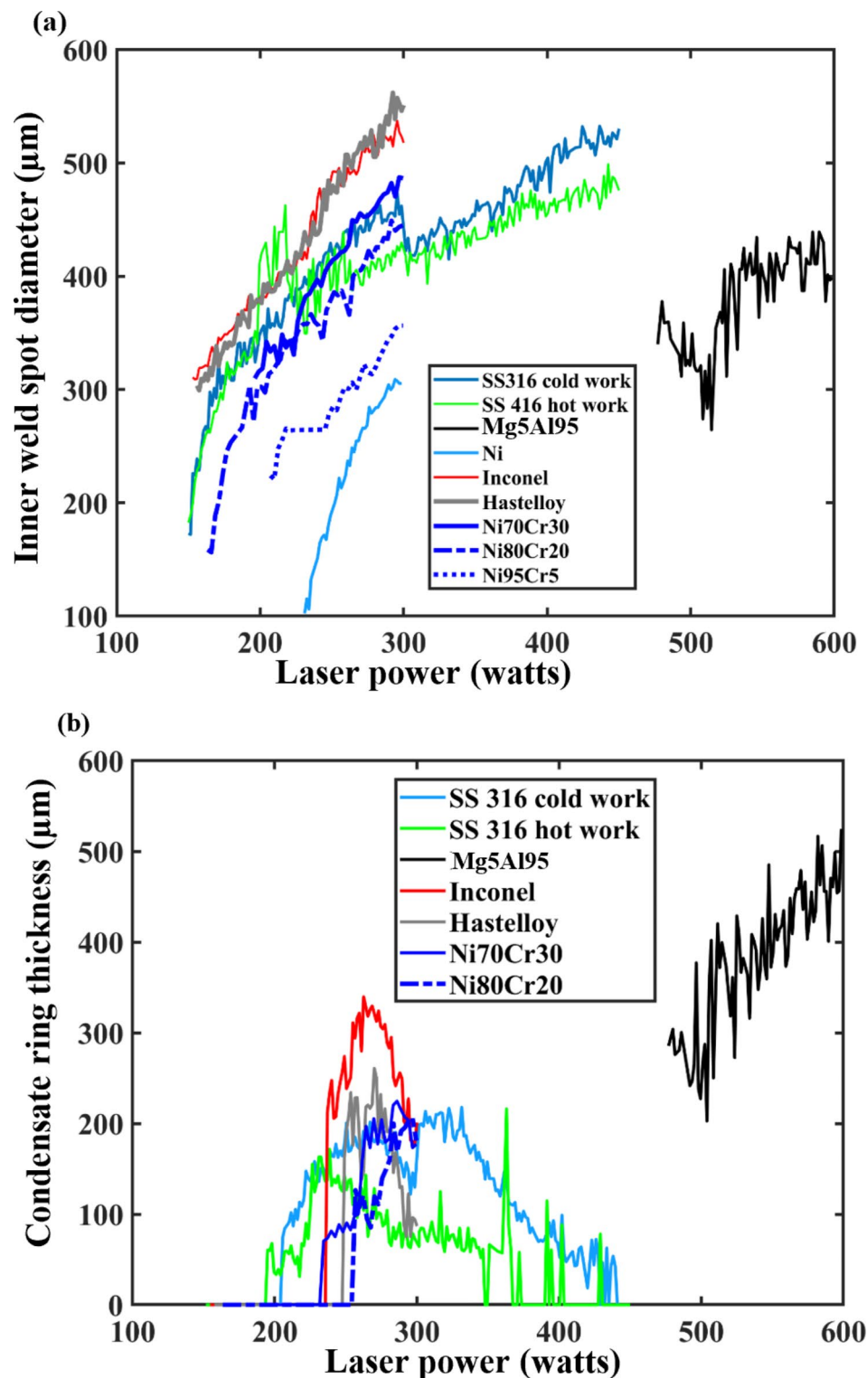


Fig. 4. (a) Inner weld spot diameter and (b) condensate ring thickness as a function of laser power for various alloys.

strong positive correlations arise primarily due to the dataset's inherent nature. The dataset lacks substantial variation in certain alloy compositions. For instance, it predominantly includes a single MgAl alloy, featuring 0.959 at% Mg and 0.041 at% Al. Consequently, this limited representation of Mg and Al variations in the dataset results in a seemingly correlated relationship between Mg and Al descriptors. A similar scenario is observed with Ti and Nb. Despite the observed strong positive correlations between certain descriptor pairs, none of them were excluded from the model training phase.

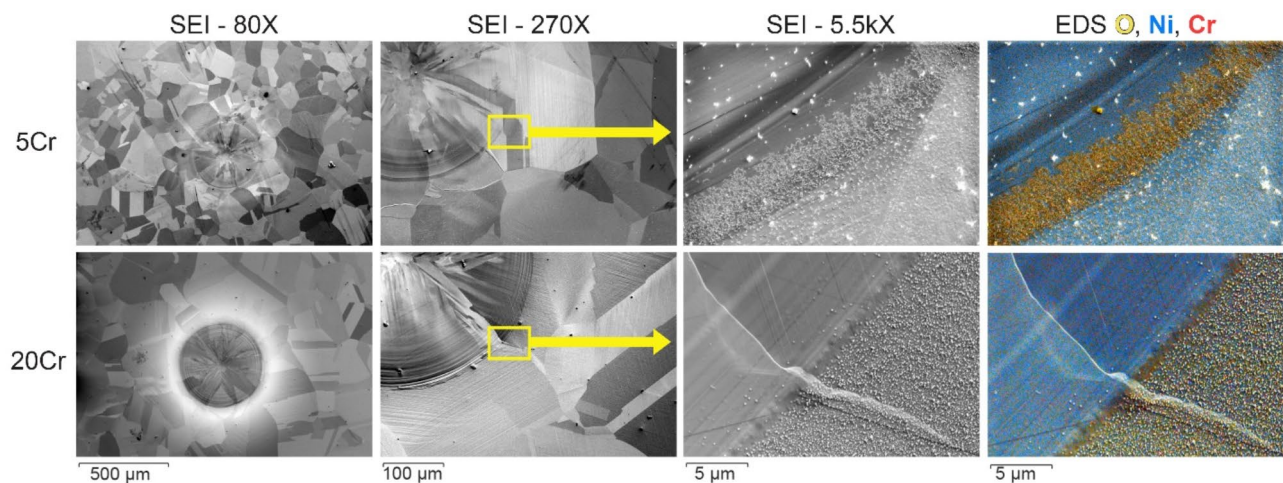


Fig. 5. SEM SEI images at variable magnification illustrate significantly increased surface condensate formation on the 20Cr sample compared to the 5Cr sample. EDS mapping confirms the surface condensate (orange coloring) is primarily chromium oxide.

This deliberate decision aims to avoid rendering the model agnostic to specific elements that might hold critical importance in yet-to-be-encountered alloys during prediction. By retaining all descriptors in the model training process, we preserve sensitivity to the potential influence of various elements, even if their correlations might be influenced by limited representation. This approach enables the model to maintain a comprehensive understanding of the diverse elemental compositions that could be present in unforeseen alloys encountered during prediction.

The distribution of elements in the training dataset is varied, with the number of rows containing each element shown in Fig. 7. From this distribution, it is evident that Nickel (Ni) and Chromium (Cr) are the most frequently occurring elements in the dataset, with 737 and 695 rows, respectively. The high occurrence of these elements suggests that the dataset contains a substantial number of Ni and Cr-based alloys, making the predictions for compositions containing these elements the most reliable. In contrast, elements like Mg, Nb, Ti, Co and W have fewer rows, indicating that predictions involving these elements might be less reliable due to the limited data available for these compositions.

The results of the model performances for the prediction of weld spot diameter and the condensate ring thickness are shown in Tables 3 and 4 respectively. The CatBoost, Gradient Boost and Random Forest Regressors produce the best performance for both the weld spot and condensate ring thickness.

Gradient Boost, CatBoost, and Random Forest are ensemble methods that construct multiple decision trees iteratively. They excel at capturing intricate non-linear relationships between alloy composition descriptors and weld characteristics, which might be the case in predicting weld spot diameter and condensate ring thickness. CatBoost, in particular, is adept at handling categorical variables present in the dataset efficiently. In situations where alloy compositions are categorical or have categorical attributes, CatBoost's handling of these variables might provide an advantage in predicting weld characteristics accurately.

The trained models serve as effective screening tools to predict materials with minimal condensate ring formation and can facilitate an inverse design approach to identify optimized compositions that minimize this phenomenon. However, in this study, the primary goal is to uncover the fundamental causes driving condensate ring formation, thereby emphasizing the importance of descriptor ranking from these models. With that goal we study the descriptor importance rankings of the three best performing models.

Discussion

In this study, we developed and validated multiple predictive models for spot weld diameter and condensate ring thickness in laser spot welding, providing new insights into the underlying factors influencing these outcomes. By employing machine learning models alongside molecular dynamics simulations, we identified the most significant descriptors governing weld quality and condensate formation.

For weld spot diameter prediction, all three models consistently identify laser power as the most influential descriptor as shown in Fig. 8 for the CatBoost model and in supplementary Figure S1 (a) and (b) for GradientBoost and Random Forest models. This aligns intuitively with its governing role in the welding process. Additionally, hardness emerges as a critical determinant, reflecting its representation of bond strengths, as cited in previous research⁴⁶. Vapor pressure attributes also garner attention from all models, rooted in fundamental scientific reasons. The significance of vapor pressure descriptors stems from the fundamental principle that higher vapor pressure metals tend to vaporize until equilibrium is achieved. Consequently, metals with higher vapor pressures exhibit increased vaporization, resulting in larger weld spot sizes. Nickel and chromium emerge prominently among the top descriptors evaluated across all three models, collectively indicating their substantial influence on condensate ring formation. Observations from Fig. 2b, f, and j corroborate these findings, demonstrating that increasing chromium content correlates with thicker condensate ring formation. This empirical evidence further

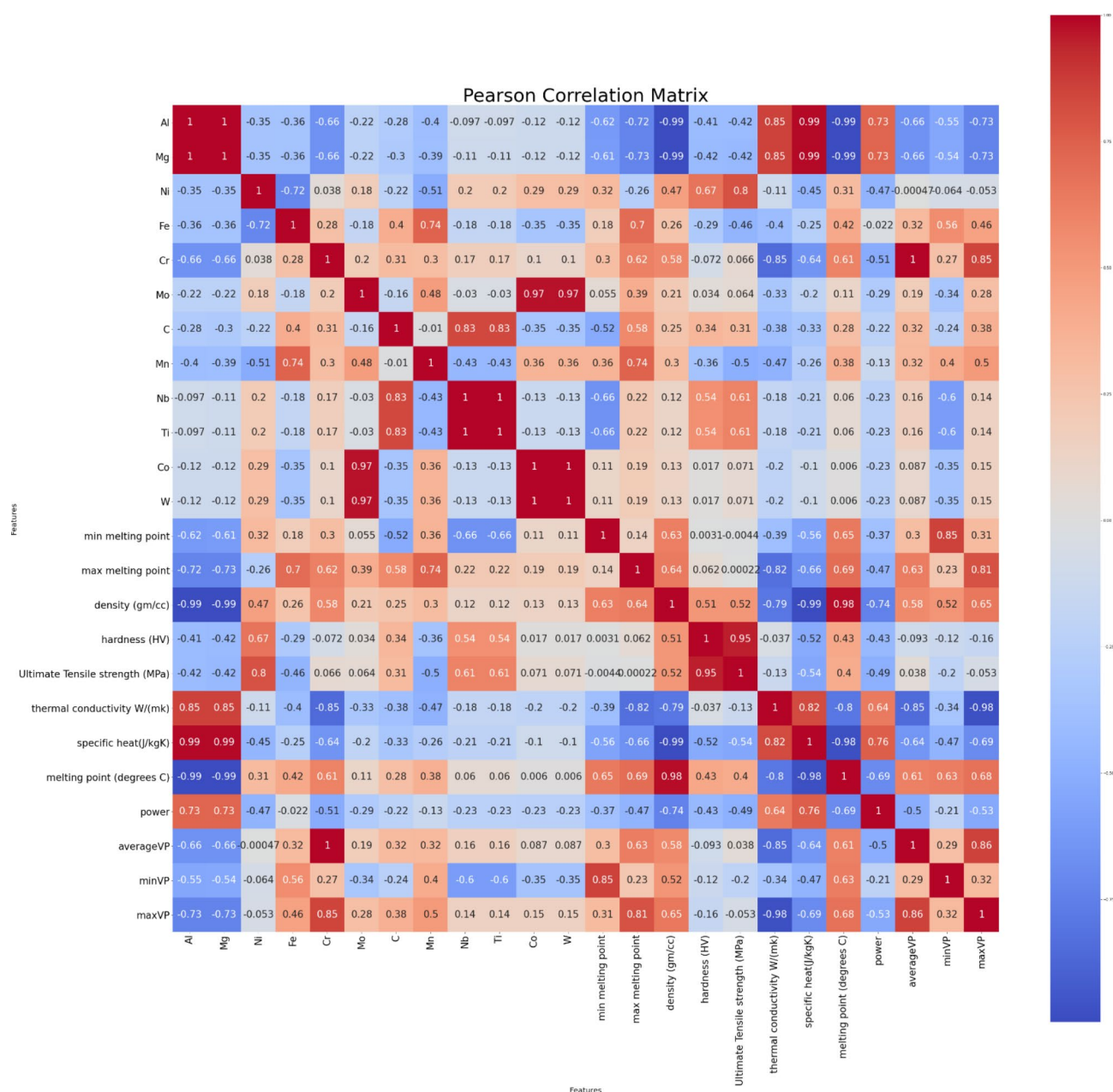


Fig. 6. Heatmap showing Pearson correlations between all pairs of descriptors in the weld spot diameter and condensate ring thickness prediction dataset. Blue and red indicate strong negative (-1) and positive (1) correlation respectively.

supports the insights gained from the machine learning models, validating the importance of alloy chemistry in influencing condensate ring thickness. These findings underscore the potential for optimizing composition to mitigate condensate ring formation in laser welding processes.

For the condensate ring thickness prediction, strikingly the laser power is not the most significant factor in any of the top three models. The convergence of findings across all three models, consistently highlighting vapor pressure and melting point descriptors as the most influential factors, stands as a corroborated result, as depicted in Fig. 9, and supplementary figure S2 (a), and (b). This result can be explained intuitively. During the welding process, the weld spot is formed as the solid substrate material goes through a phase change forming a liquid melt pool. While the melting process is driven primarily by heating (ultimately weld power), subsequent formation of condensate requires vaporization. Thus, vapor pressure emerges as the most critical factor in controlling condensate ring thickness.

The condensate ring thickness prediction models rank the vapor pressure attributes at ~40% importance in all three models thereby strongly indicating it as a primary factor in the formation of the condensate ring. While feature importance can indicate which features significantly impact the target variable, it does not provide information on whether an increase in these features will result in an increase or decrease in the target value.

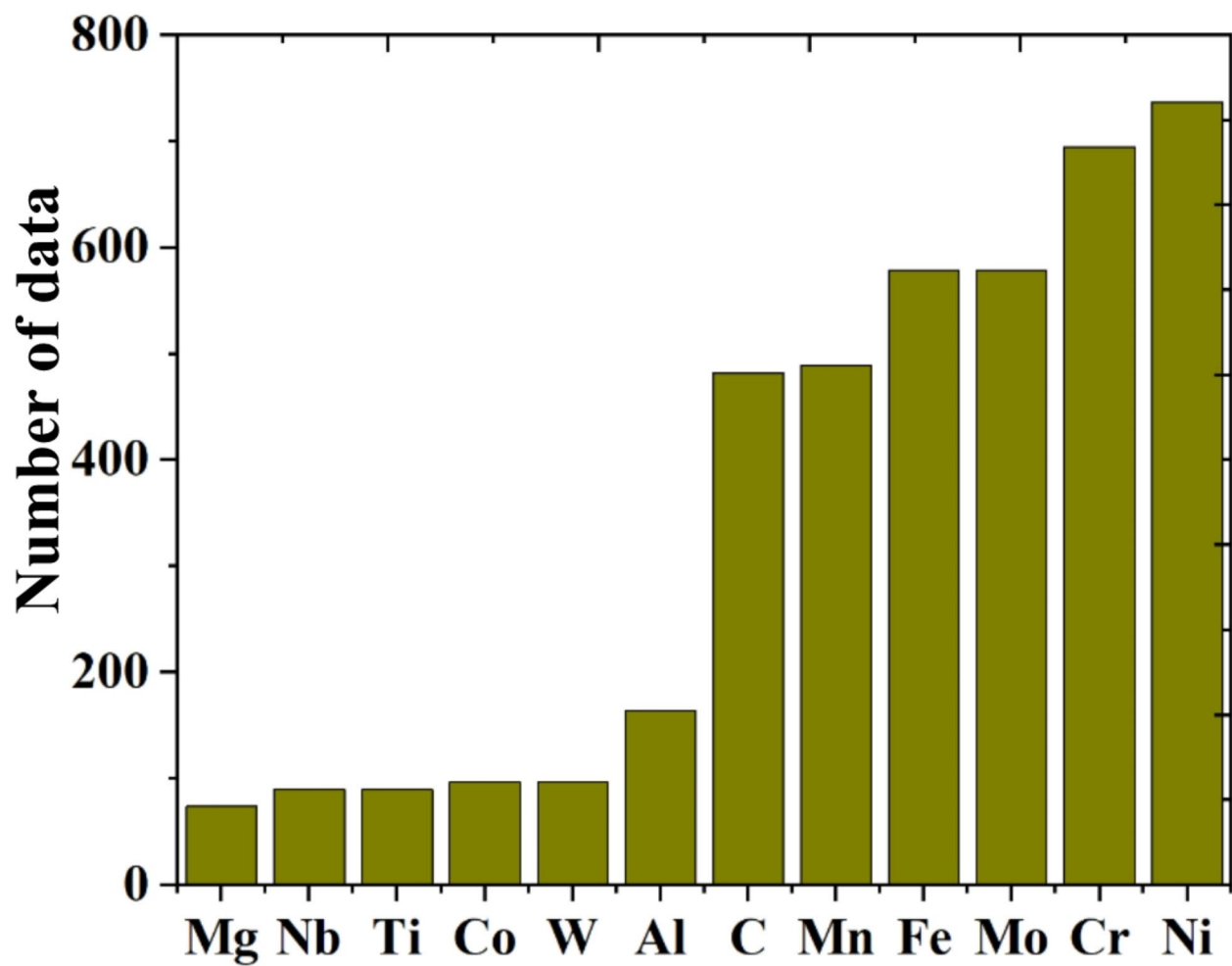


Fig. 7. Distribution of the elements over the dataset.

Model	R2_score	parameters
CatBoosting Regressor	0.9859	Learning rate (LR): 0.03, depth: 6, iterations: 1000, loss function: RMSE
Gradient Boost Regressor	0.9792	Loss: squared error, LR: 0.1, n_estimators: 100
Random Forest Regressor	0.9778	N_estimators: 100, criterion: squared error
XGBRegressor	0.9714	LR:0.3, n_estimators: 100, max_depth: 6
Decision Tree	0.9616	Criterion: squared error, max_depth: None
Gaussian Process Regressor	0.9360	Kernel: Radial Basis Function, alpha: 1e-10
AdaBoost Regressor	0.8854	Base_estimator: DecisionTree with max_depth= 3, n_estimators: 50, LR: 1
Linear Regression	0.8582	Fit_intercept: true
Ridge	0.3791	Alpha: 1
Lasso	0.3342	Alpha: 1, max_iter: 1000

Table 3. Weld spot diameter prediction performance of various models.

To capture this mathematical dependence and understand the specific contributions of each feature, polynomial regression fitting was employed.

The polynomial regression model attempts to predict the numerical value of the condensate ring thickness by capturing the complex interactions between the various input descriptors. All descriptors have been normalized with values between 0 and 1 that eliminates the problem of encountering dimensional inaccuracies to construct a predictive polynomial equation. This approach allows us to quantify how changes in hardness, vapor pressure, laser power, and melting point, among other factors, influence the formation of the condensate ring. Using the

Model	R2_score
CatBoosting Regressor	0.9835
Gradient Boost Regressor	0.9802
Random Forest Regressor	0.9751
XGBRegressor	0.9727
Decision Tree	0.9680
PolynomialFeatures	0.9651
AdaBoost Regressor	0.9649
Linear Regression	0.9233
Ridge	0.9106
Lasso	0.9089

Table 4. Condensate ring thickness prediction performance of various models.

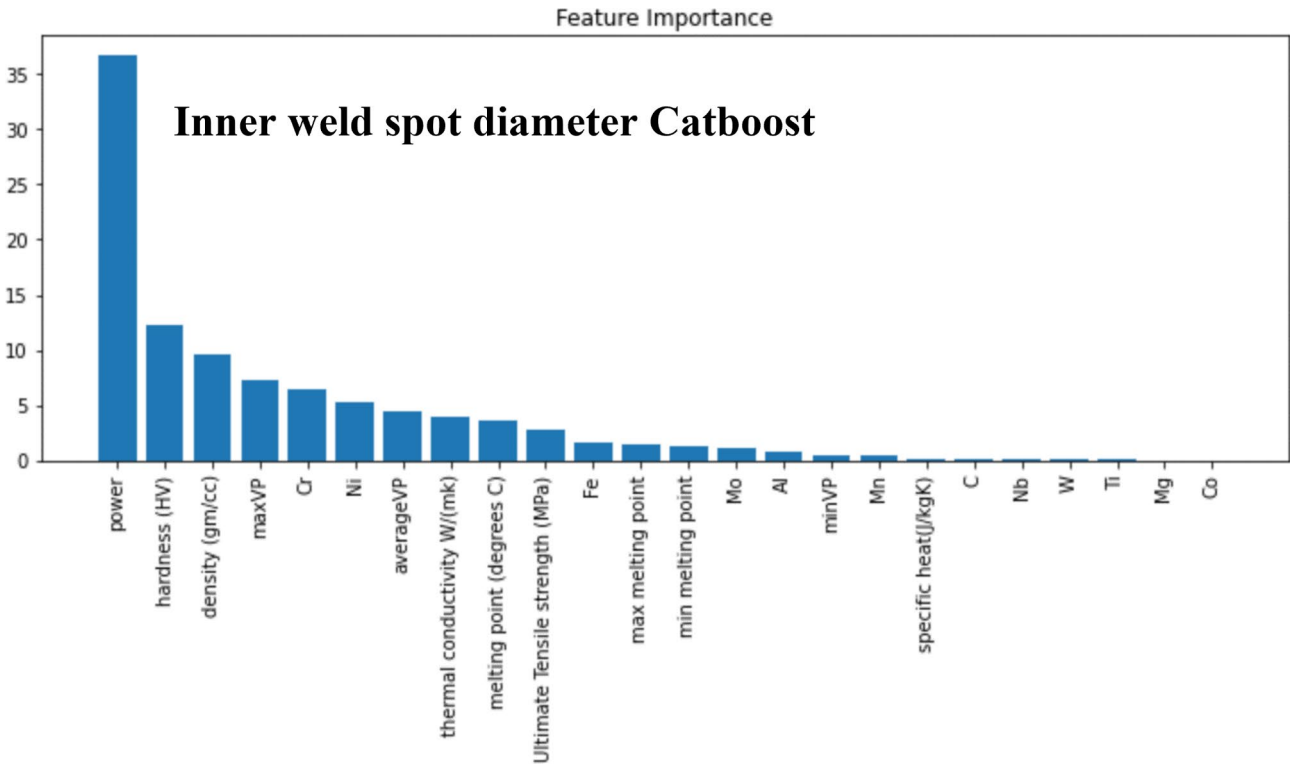


Fig. 8. Descriptor importance ranks for weld spot diameter prediction from the CatBoost model.

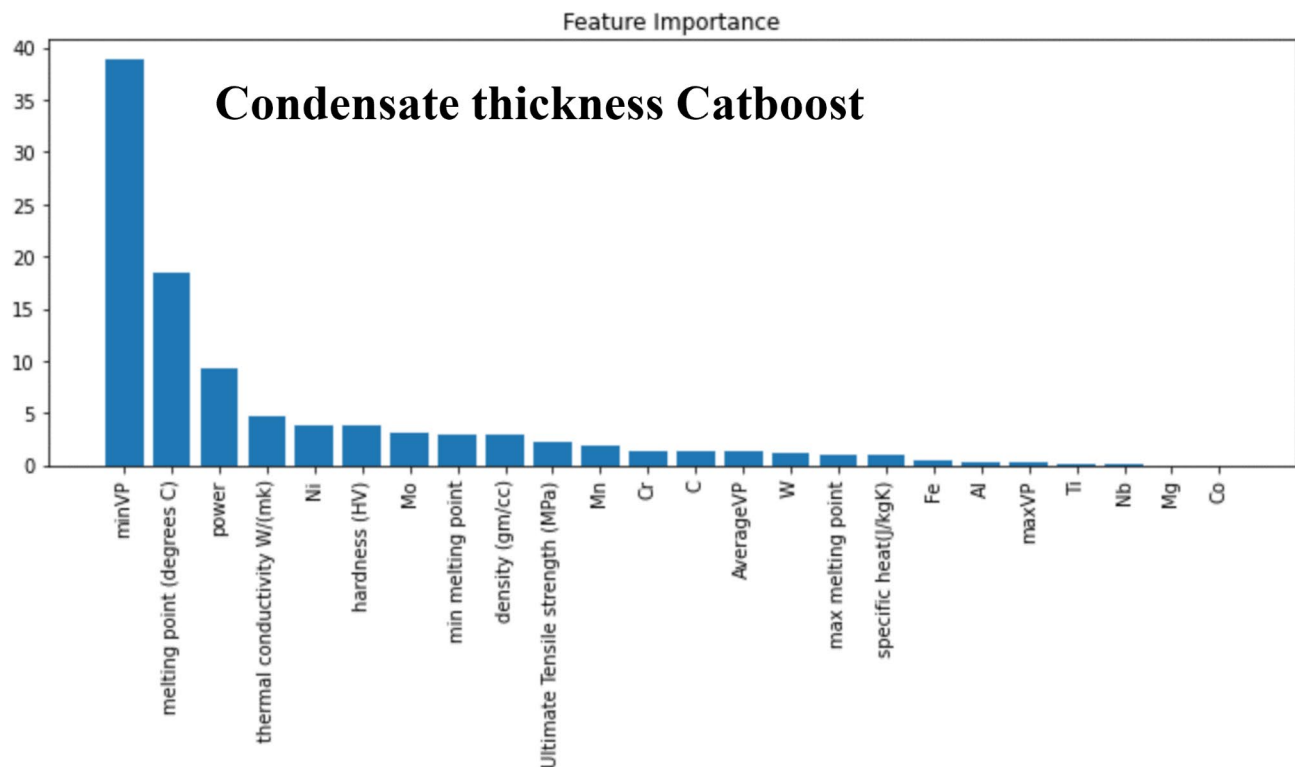


Fig. 9. Descriptor importance ranks for condensate ring thickness prediction from the CatBoost model.

top 5 features namely minimum vapor pressure, average vapor pressure, hardness, power and melting point, a polynomial regression analysis was conducted. The polynomial regression yielded a formula of degree 2, for predicting condensate ring thickness, incorporating the mentioned top descriptors:

$$\begin{aligned}
 \text{condensate ring thickness } (\mu m) = & 8.58 \times 10^{11} + 4.05 \times 10^{14} \times \text{hardness} \\
 & - 6.28 \times 10^{13} \times \text{Average VP} + 1.86 \\
 & \times 10^3 \times \text{power} - 2.5 \times 10^{13} \times \text{minVP} \\
 & - 1.01 \times 10^{14} \times \text{melting point} + 2.79 \\
 & \times 10^{11} \times \text{hardness}^2 + 2.77 \times 10^{14} \\
 & \times \text{hardness} \times \text{average VP} + 1.51 \times 10^3 \\
 & \times \text{hardness} \times \text{power} + 3.28 \times 10^{14} \\
 & \times \text{hardness} \times \text{minVP} - 1.08 \times 10^{15} \\
 & \times \text{hardness} \times \text{melting point} + 5.9 \times 10^{12} \\
 & \times \text{average VP}^2 + 1.86 \times 10^3 \\
 & \times \text{average VP} \times \text{power} - 1.08 \times 10^{14} \\
 & \times \text{average VP} \times \text{minVP} + 1.01 \times 10^{14} \\
 & \times \text{average VP} \times \text{melting point} - 6.04 \\
 & \times 10^2 \times \text{power}^2 + 1.14 \times 10^3 \times \text{power} \\
 & \times \text{minVP} - 4.28 \times 10^3 \times \text{power} \times \text{melting point} \\
 & + 1.02 \times 10^{14} \times \text{minVP}^2 - 1.61 \times 10^{14} \\
 & \times \text{minVP} \times \text{melting point} + 2.52 \times 10^{14} \times \text{melting point}^2
 \end{aligned}$$

In the above formula most terms have a positive correlation. For example the power term with degree one has a positive coefficient of 1.86×10^3 indicating that increasing the power will cause a thicker condensate ring to be formed, however the power² term has a negative (-6.04×10^2) correlation indicating a diminishing return effect where very high values of power and melting point can lead to a reduction in ring thickness. This is consistent with experimental observations where most materials exhibit a smaller condensate ring thickness at higher powers (Fig. 4b), suggesting that the formula mimics reality up to a certain extent. The polynomial regression model provides a nuanced understanding of how different factors interact to influence condensate

ring formation. This mathematical dependence allows for more precise adjustments in the welding process to optimize outcomes.

The characteristic trend where the condensate ring thickness initially increases with power, reaches a maximum, and subsequently decreases at higher power levels is conjectured to be driven by the dynamic interplay between vaporization and condensation processes during laser-material interaction. At lower laser power levels, the absorbed energy leads to a gradual increase in local heating and vaporization. As the material vaporizes, the vapor cools and condenses around the weld zone, leading to the formation of a thicker condensate ring. As the power increases further, the vaporization rate reaches a maximum, resulting in the thickest condensate ring, as the balance between vapor generation and condensation is optimized. However, beyond this point, at higher power levels, the formation of vapor plumes over the weld pool becomes more significant. These phenomena can deflect or absorb laser energy, reducing the effective energy absorbed by the material surface and, consequently, decreasing the amount of vapor available for condensation⁵. Increased vapor ejection velocities at higher powers may lead to the dispersal of vaporized material away from the immediate weld zone, further reducing the local condensation. As a result, the condensate ring thickness decreases at these elevated power levels.

To corroborate the relationship between vapor pressure and condensate ring formation, MD simulations were performed mimicking the spot welding process. The simulations were done on Ni70Cr30, Ni80Cr20 and Ni100Cr00 series, to study the effect of varying chemical composition on the condensate ring formation and identify the role of vapor pressure. Figure 10 shows the results of simulations on these alloys where Fig. 10a to e show the temporal progression of material vaporization from the surface into the vacuum for a representative Ni70Cr30 alloy. As the material absorbs additional heat with every passing time step, some of the atoms on the surface accumulate enough energy to transition into the liquid phase and then vaporize into the vacuum region. Figure 10f–i show the temporal depiction of the vertical column into which heat is added every time step to simulate a laser spot. Atoms absorb the incident energy and escape outside the bounds of the column dimension until the column volume is devoid of any atom inside it. During this process, the atoms leaving the surface and vaporizing into the vacuum region are tracked. These atoms are the ones that will eventually condense to form the condensate ring. The higher the number of the atoms vaporized, the thicker is the resulting condensate ring.

The count of atoms leaving both the surfaces and vaporizing into the vacuum region is plotted in Fig. 11a for the three alloys. Clearly, addition of Cr correlates with a higher number of vaporized atoms. This can be attributed to the higher vapor pressure of Cr than Ni, where our calculations yield a value of ~ 458 Pa for vapor pressure of Cr at its melting point 2130 K and a vapor pressure of 0.44 Pa for Ni at its melting point 1726 K. Confirmation is obtained from monitoring the pressure of the simulation as shown in Fig. 11b.

Figure 11b shows the variation of total pressure in the system as a function of time. Initially, before the vaporization of atoms begins from surface, the vacuum above and below the volume of the bulk material results in negative pressure at $t = 0$. With the progression of time, the total pressure increases due to the formation of vapor phase. The total pressure is comprised of the sum of partial pressures of all the gaseous components in a system and can be expressed as $P_{total} = \sum_i P_i X_i$, where P_i and X_i are the vapor pressure and mole fraction of substance i . The pressure in Ni70Cr30 is the highest (black line) followed by Ni80Cr20 and then pure Ni. Clearly, it aligns with our previous hypothesis and experimentation observation in Fig. 5, that addition of Cr into the system raises the vapor pressure, thus increasing the number of atoms vaporizing from the surface until the vapor pressure has been met at equilibrium.

In a study by Chang and Na⁴⁷, a combined approach using the Finite Element Method (FEM) and a neural network was employed to predict the bead shape in laser spot welding of AISI type 304 stainless steel. This hybrid modeling technique effectively addressed the limitations of each individual method, providing accurate predictions of weld characteristics. The study emphasized the critical role of temperature-dependent thermal properties and laser parameters such as pulse energy, pulse duration, and focal position in determining the weld bead shape. Similar to these findings, our research highlights the importance of vapor pressure, melting point descriptors, and other thermal properties in governing condensate ring formation.

By leveraging machine learning models, we can accurately predict the condensate ring thickness, drawing parallels with the neural network's effectiveness in predicting weld bead shapes from⁴⁷. Furthermore, our machine learning predictions have been validated by MD simulations. These simulations corroborate the machine learning findings by elucidating the vaporization dynamics and confirming the significant role of vapor pressure in surface vaporization. The validation of numerical models with experimental results, underscores the significance of empirical data in supporting simulation outcomes. In our research, experimental observations using SEM and EDS have been employed to validate both the machine learning and MD simulation results, enhancing the credibility of our findings.

Conclusions

The fundamental factors leading to the formation of condensate rings in laser spot welding process were investigated using experimental observations reasoned by machine learning insights and MD simulations. ML models were trained on collected experimental data using chemical composition descriptors, physical quantities of the alloys and process parameters to predict weld spot diameter and condensate ring thickness on various alloys. The models revealed that condensate ring formation is primarily governed by the vapor pressure of the metals in the alloys. Higher vapor pressure of the constituent metals facilitates increased vaporization of atoms from the melt pool resulting in formation of a thicker condensate ring. This hypothesis was confirmed by MD simulations on Ni70Cr30, Ni80Cr20 and pure Ni where Ni70Cr30 exhibited highest number of atoms vaporizing from the surface and pure Ni exhibited the least owing to the high vapor pressure of Cr (~ 458 Pa) and low vapor pressure of Ni (0.44 Pa). These insights contribute significantly to the understanding of underlying mechanisms driving condensate ring formation in welding processes, emphasizing the crucial role of vapor pressure as a governing factor in such phenomena.

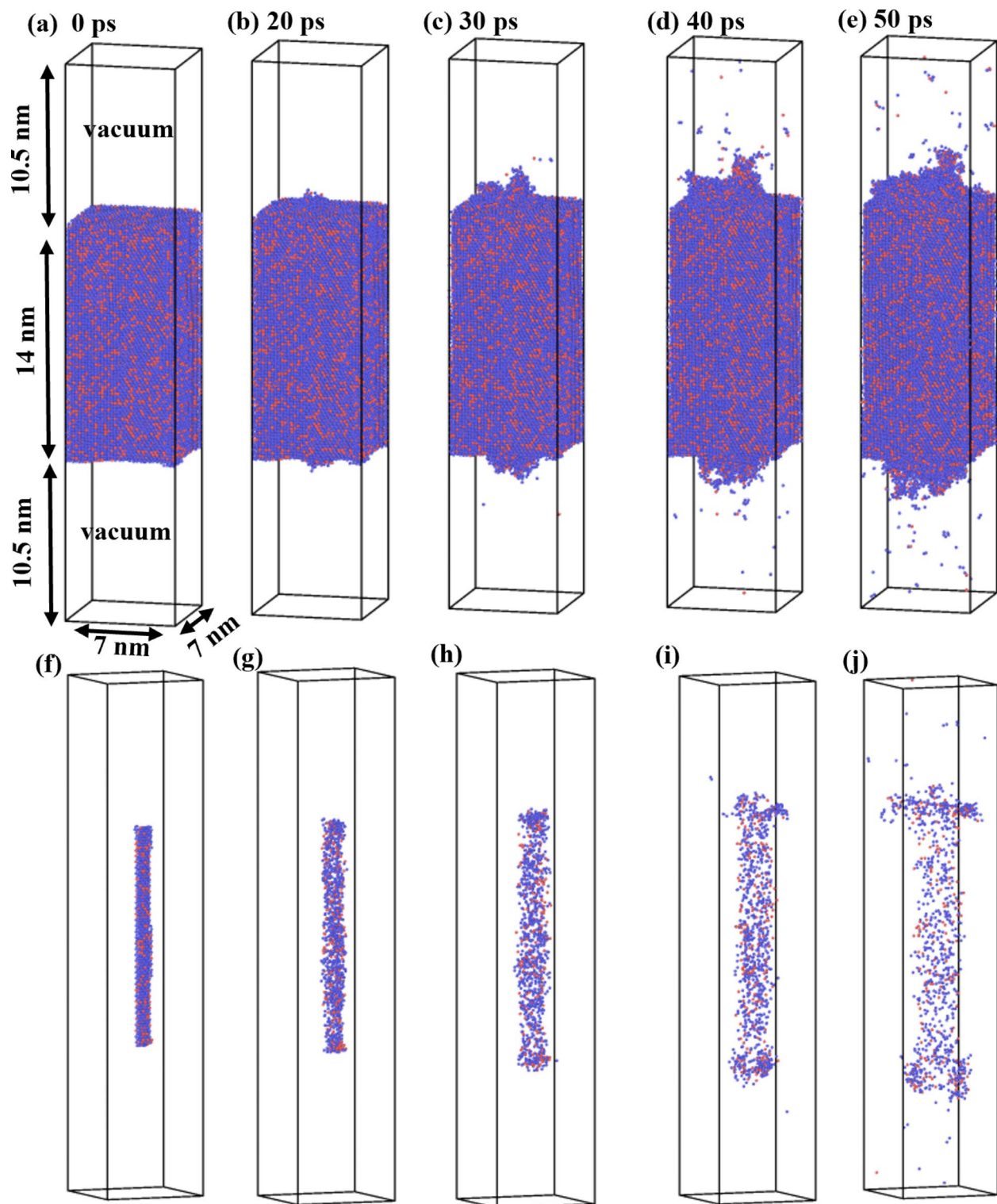


Fig. 10. Molecular dynamics model of the laser spot welding process on Ni70Cr30 alloy at (a) $t = 0$ ps, (b) $t = 20$ ps, (c) $t = 30$ ps, (d) $t = 40$ ps and (e) $t = 50$ ps depicting the vaporization of atoms from the surface into the vacuum region. The column heated by the laser is shown at (f) $t = 0$ ps, (g) $t = 20$ ps, (h) $t = 30$ ps, (i) $t = 40$ ps and (j) $t = 50$ ps. Melting takes place until all the atoms in this column escape outside the bounds of the cuboidal volume, rendering the column volume devoid of particles.

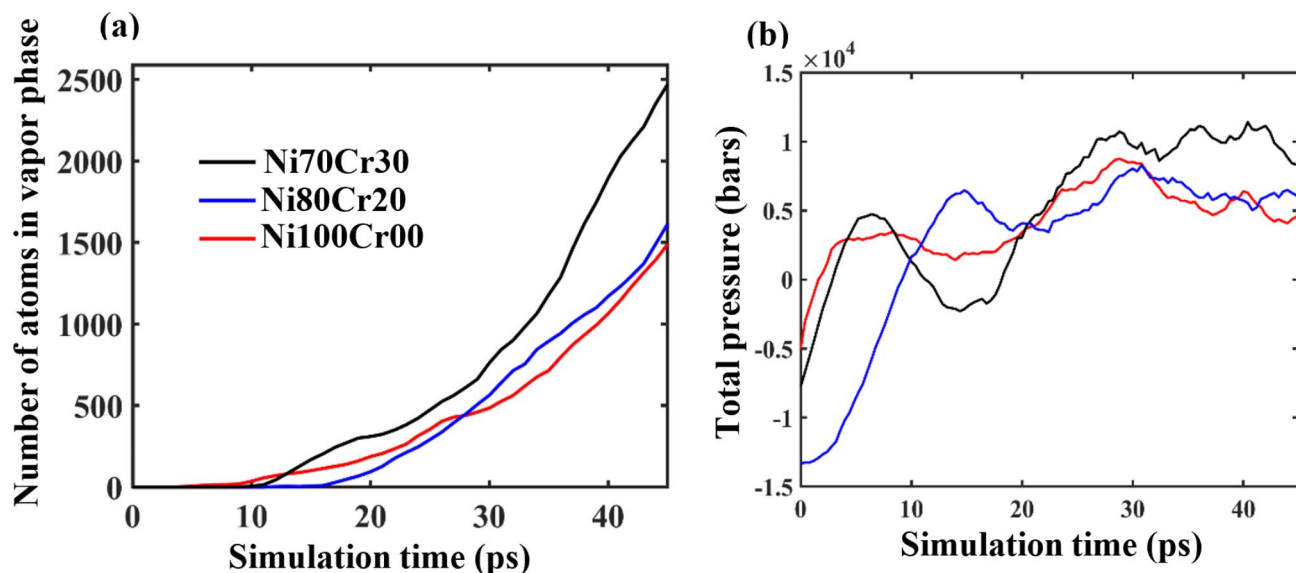


Fig. 11. (a) Temporal variation of number of atoms vaporizing from the surface of Ni70Cr30 (black), Ni80Cr20 (blue) and Ni100Cr00 (red). (b) Total pressure in the system as a function of time, during the laser spot welding simulation for the same alloys.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 28 September 2024; Accepted: 12 November 2024

Published online: 03 December 2024

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Acknowledgements

This research was supported by the AT SCALE Initiative under the Laboratory Directed Research and Development (LDRD) program at Pacific Northwest National Laboratory (PNNL). PNNL is a multiprogram national laboratory operated for the U.S. Department of Energy (DOE) by Battelle Memorial Institute under Contract No. DE-AC05-76RL01830. The support by Anthony Guzman and Irving Brown is acknowledged for performing metallographic sample preparation.

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A.R.: conceptualization, investigation, visualization, writing—original draft, writing—review & editing. L.H.: investigation, writing—review & editing. N.R.O.: writing—review & editing. K.R.F.: investigation, writing—review & editing. D.H.: writing—review & editing. F.H.: writing—review & editing. M.L.T.: writing—review & editing, resources, funding acquisition. D.K.S.: writing—review & editing, resources, funding acquisition, M.J.O.: writing—review & editing, supervision.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-79755-8>.

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